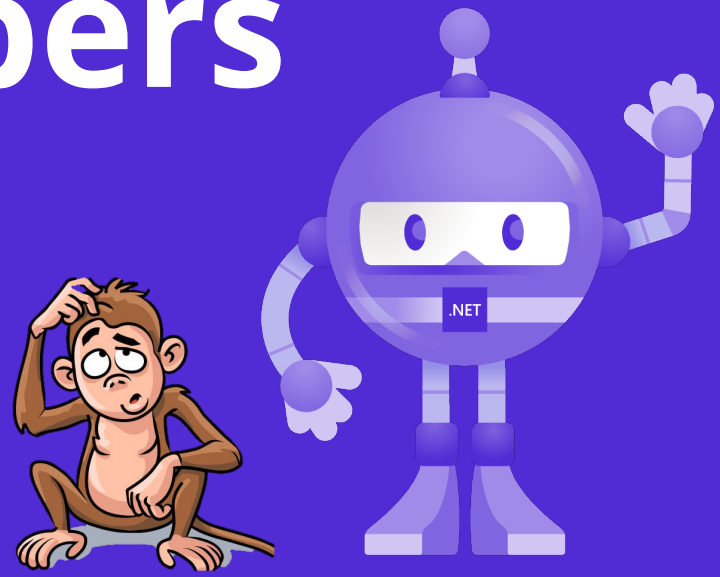
.NET MAUI
for Xamarin Developers

David Ortinau
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@davidortinau





David Ortinau

@davidortinau

24 yrs. web, interactive, mobile

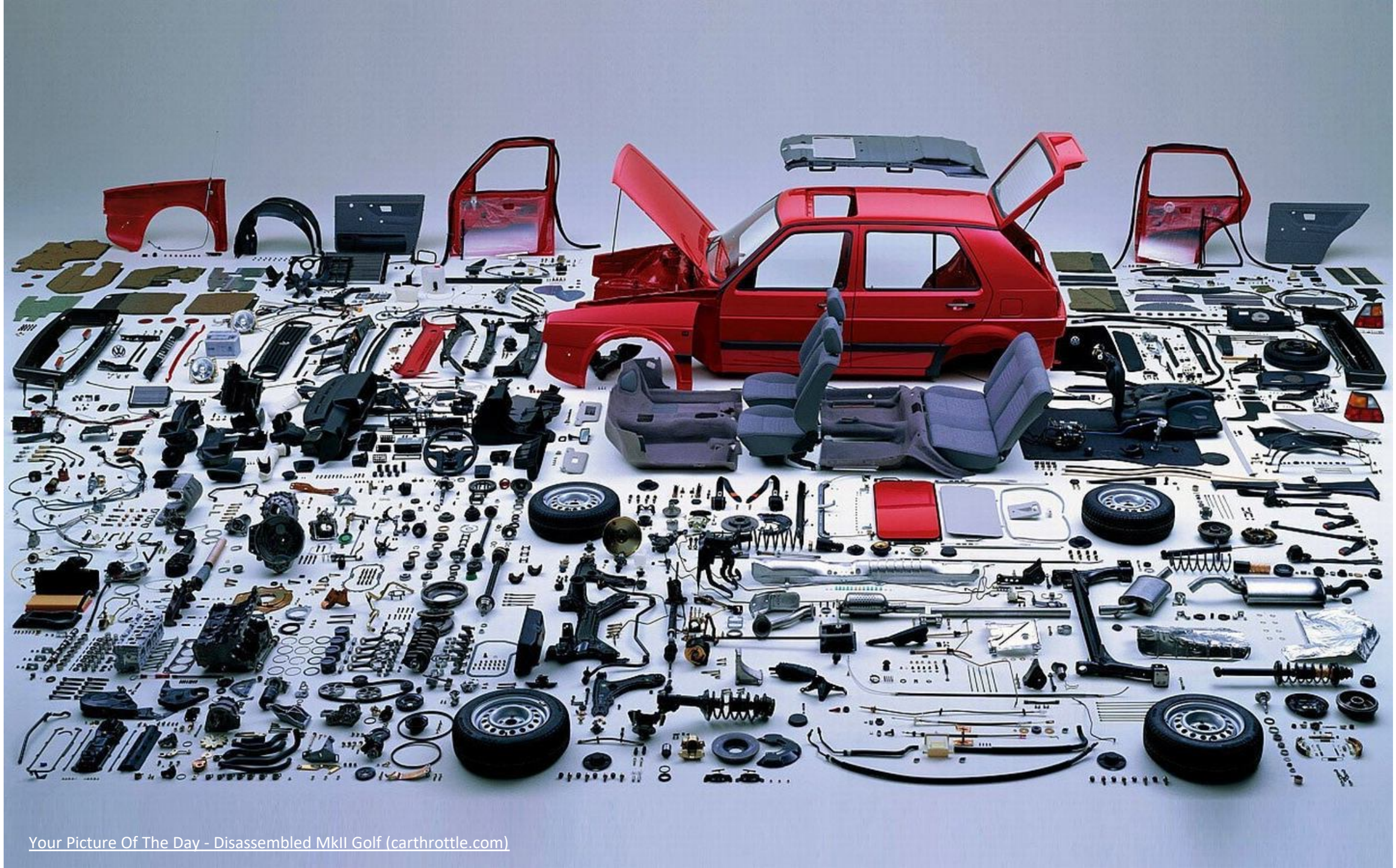
Xamarin MVP 2013 - 2017

BA English, Maryville University

Principal Program Manager
~~Visual Studio Mobile Developer Tools~~
~~Xamarin~~
.NET MAUI

Microsoft

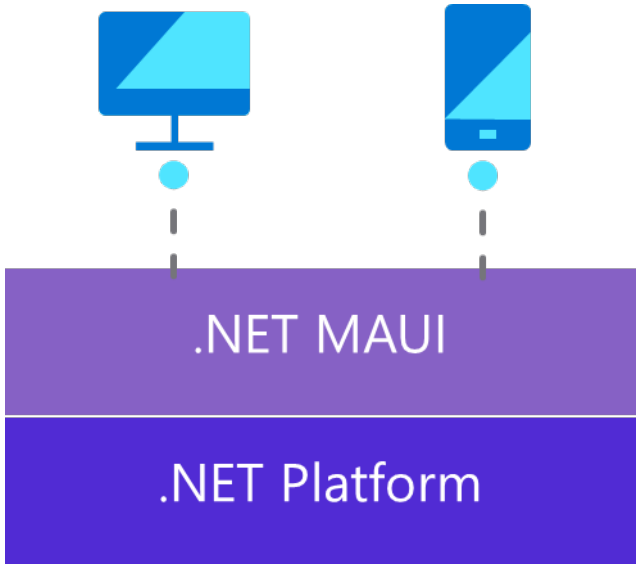
.NET MAUI is the evolution of Xamarin.Forms extended from mobile to desktop scenarios with UI controls rebuilt from the ground up for performance and extensibility.



Your Picture Of The Day - Disassembled MkII Golf (carthrottle.com)

.NET Multi-platform App UI (MAUI)

- Xamarin integrated into .NET
  - github.com/dotnet/maui
  - github.com/dotnet/android
  - github.com/dotnet/macios
- Common .NET experiences
  - CLI
  - Visual Studio Code
- Android, iOS, macOS, and Windows
- Support for modern app models like Blazor and Comet



Goals of .NET MAUI

Quality

12 months of Xamarin quality releases

Reorganized project structure.

Standardized class naming and structure

Performance

Cold start <= 1000ms

Responsive UI rendering, animations, and transitions

Reduced view nesting

Design

Simple and elegant default UI theme

Easy control extensibility

Quick cross-platform templating

Platforms

Android

iOS

macOS

Windows

	Xamarin.Forms 5	.NET MAUI
Platforms		
Android	API 19+	API 21+
iOS	9 – 15	10+
Linux (GTK#)	Community	Community
macOS	Community	Microsoft
Tizen	Samsung	Samsung
Windows	UWP Microsoft WPF Community	WinUI 3 Microsoft WPF Community
Features		
Renderers	Tightly coupled to BindableObject	Loosely coupled, no Core deps
App Models	MVVM / RxUI	MVVM / RxUI (MVU experimental)
Single Project	No	Yes
Multi-Targeting	No	Yes
Multi-Window	No	Yes
Pixel Drawn Controls	No	Yes
Misc		
.NET	X.iOS, X.Android, UWP, …	.NET 6 +
Project System	Franken-Proj	SDK Style
dotnet CLI	No	Yes
Tools		
Visual Studio 2019	Yes	Yes
Visual Studio 2019 for Mac	Yes	Yes
Visual Studio Code	No	Experimental

.NET MAUI SDK Timeline

Q4 2020 – Q1 2021

Feature	Description	Status
Namespace changes to System.Maui	Xamarin.Forms will become System.Maui	Deferred
Handlers: Containers, Layouts, Views	Hello performance and extensibility	In Progress
Remove Obsoletes	Housekeeping	In Progress
Host Builder	.NET unified experiences	In Progress
Remove Data Pages	You use it? Probably not. Why is it still there?	Completed
Remove Themes	See Data Pages	Completed
Startup registry	No more reflection	Completed

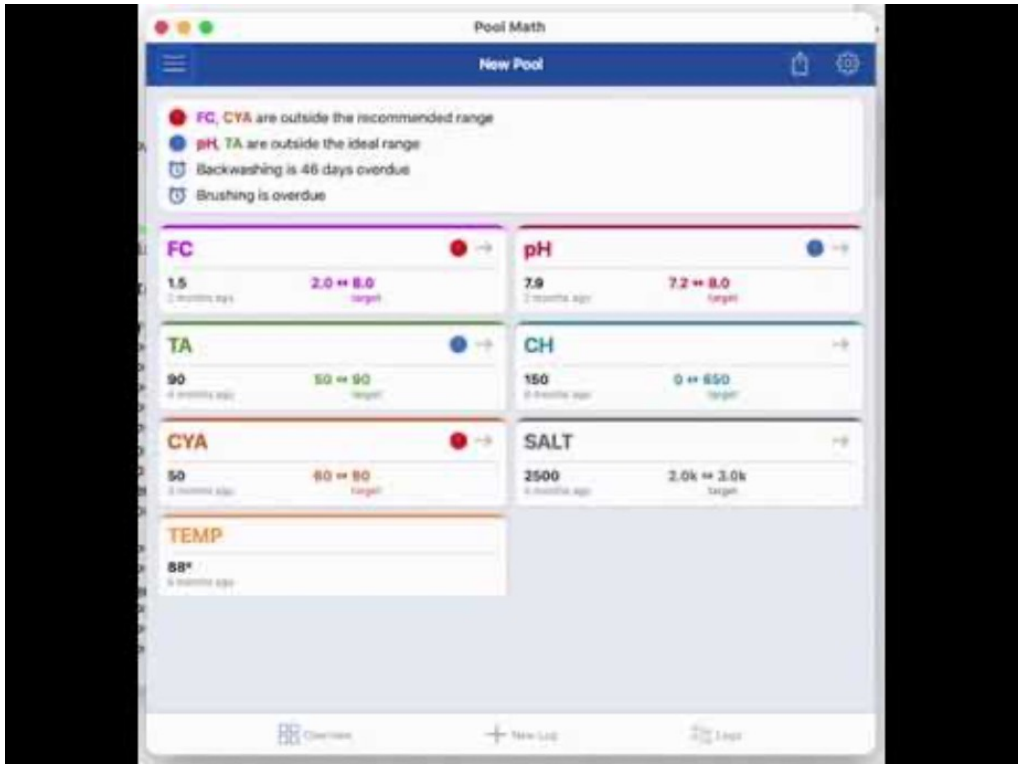
.NET MAUI SDK Timeline

Q2 2021 – Q3 2021

Feature	Description	Status
Intro Cross-platform Lifecycle		Committed
Move “AndExpand” layout option	Belongs only on StackLayouts	Committed
Width/Height	Deprecate “Request” props, update behavior	Committed
Merge Essentials	Remove overlapping APIs	Committed
Improved native images	GlideX	Committed
Housekeeping	Ianimatable, Behavior/Trigger inheritance	Committed
macOS	Catalyst	Committed
Windows	WinUI 3 pending GA	Committed

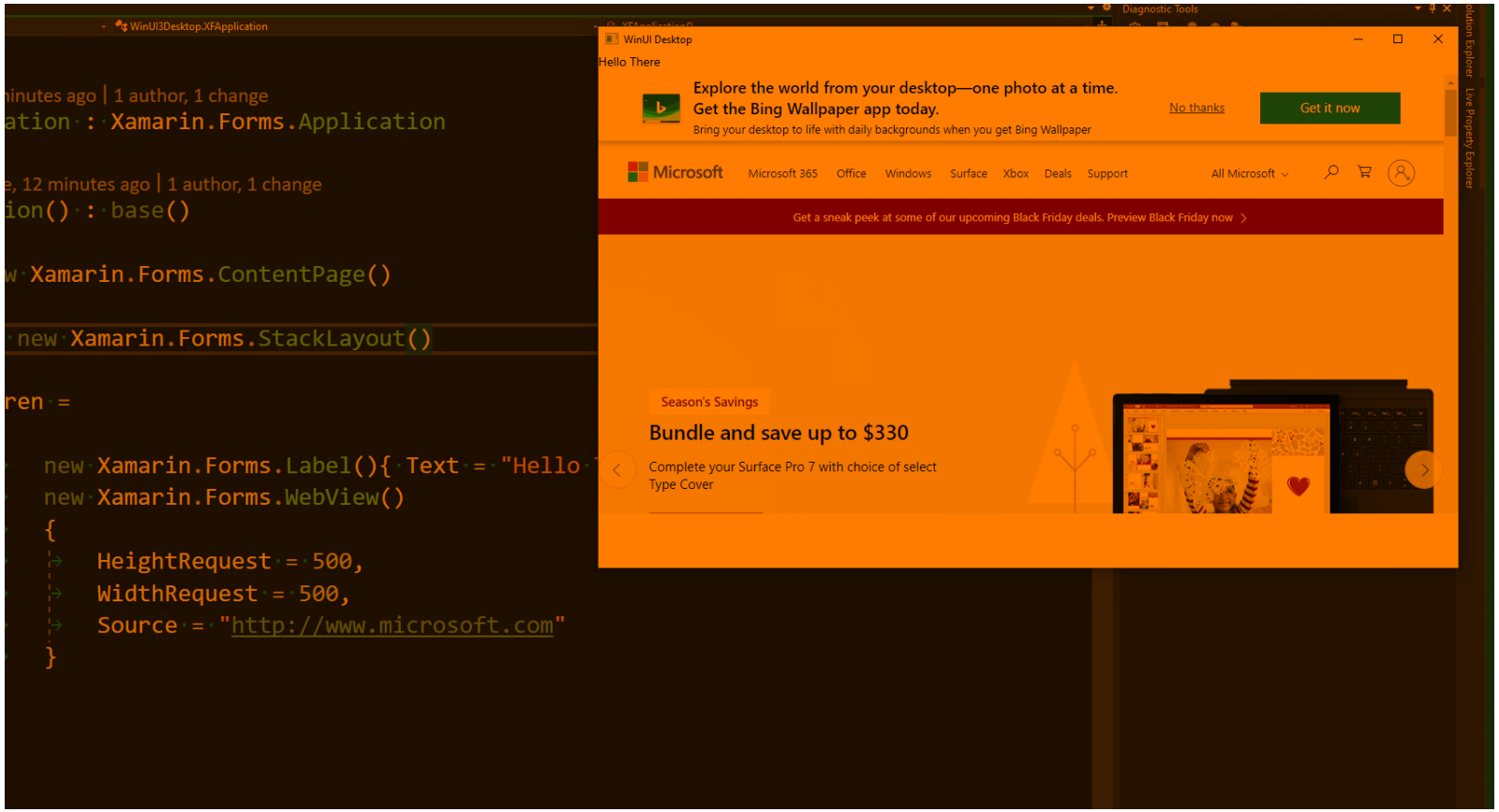
Releases in sync with .NET 6 Previews
November 2021 GA

Mac Catalyst



Jonathan Dick – Pool Math
Thanks Frank! https://praeclarum.org/2020/11/10/catalyst.html

WinUI 3



https://github.com/xamarin/Xamarin.Forms/tree/winui3_desktop

Installing .NET MAUI

From the beginning

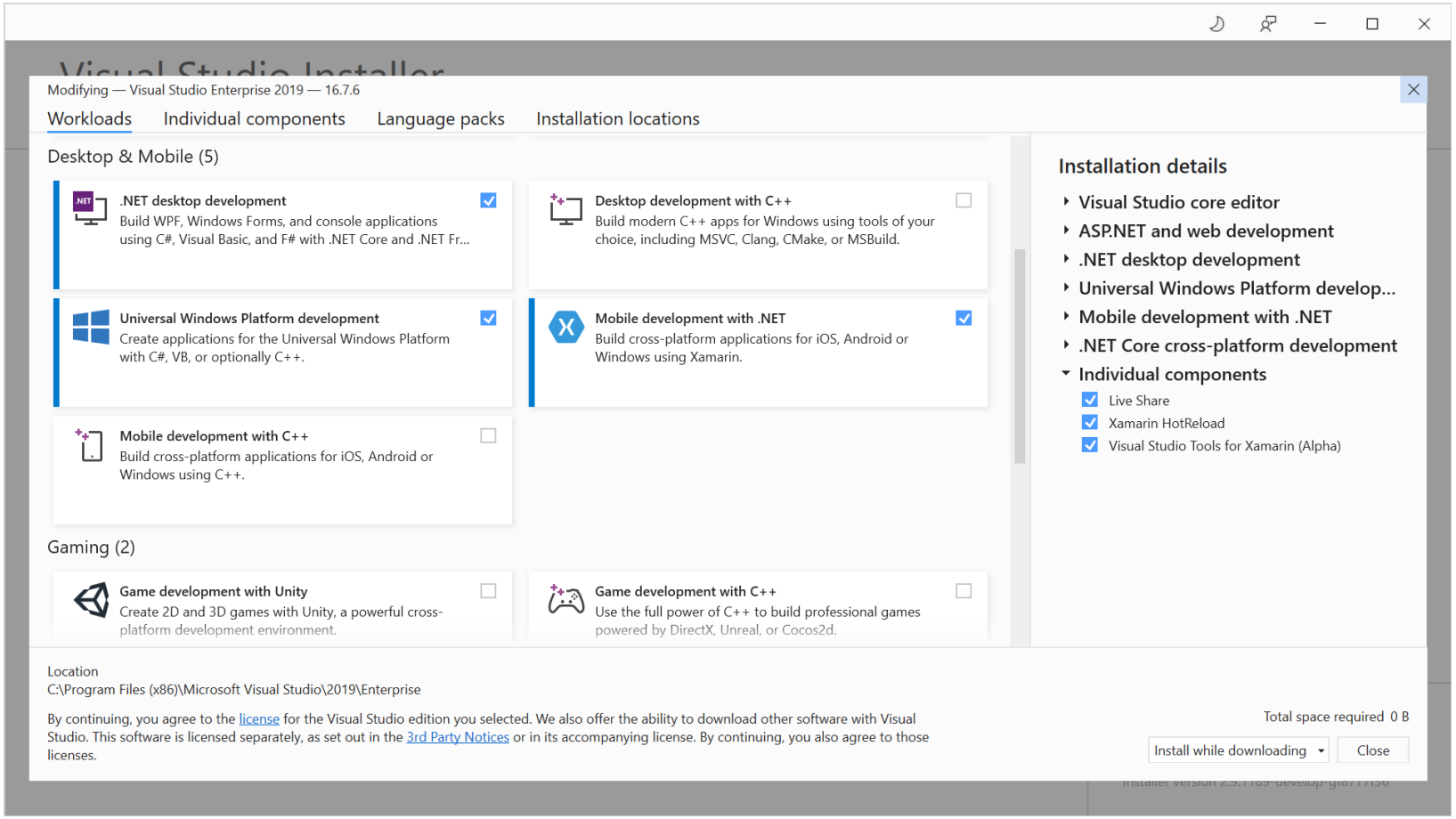
.NET Installer

- Optional workload packs (design proposal)
  - Android, iOS, macOS, WPF, etc.
  - .NET MAUI

- SDKs are decoupled from tooling
- Side by side SDK support for OS releases
- Ship with .NET
- Hosted as NuGet SDKs

```
dotnet install maui
```

Visual Studio Installer



Migrating to .NET MAUI

When you say “easy” what exactly do you think is easy?!?!

Moving to SDK Style projects

- try-convert
  - Tool will make a best effort to convert your csproj to SDK Style
  - Update target framework monikers
  - Run from CLI
  - Investigating IDE integration

Targeting .NET MAUI



- Namespaces
  - try-convert or similar will be implemented to move to System.Maui
- Custom Renderers
  - Recommendations
    - Remove those you no longer need
    - Replace the rest with handlers
  - Use built-in adapter shim and Xamarin.Forms.Legacy
- Build and fix
  - Deprecated APIs
  - Fixed APIs

Xamarin.Forms.Legacy

This is a (yet to exist) NuGet package where we can put things that you'll need to maintain compatibility for Xamarin.Forms "things" until you can complete any migration to .NET MAUI.

- Compatibility package
- Renderers
- Obsoleted

Third-Party Ecosystem

- Control Vendors & Third-Party Libraries
  - Recommendations
    - Update to .NET MAUI optimized packages
    - Recompile and target .NET 6 TFMs
  - Use adapter-shim and Xamarin.Forms.Legacy

- Framework Libraries (Prism, FreshMvvm, MvvmCross, RxUI...)
  - Recommendations
    - Update to .NET MAUI optimized packages

- We host a monthly sync call. Open invitation. 

How long is the transition period?

- November 2020
  - Announcements and Plans begin
  - Begin building migration help and producing guidance
- November 2021
  - Xamarin.Forms 5.x with latest iOS and Android support
  - .NET MAUI GA
- November 2022
  - Xamarin.Forms end of support
  - .NET MAUI vNext

Preparing for .NET MAUI

What should you do today to make for an easy migration

DO

Use AppCompat on Android

DO

Use Fast Renderers

DO NOT

Freak out



DO

Remove unnecessary custom renderers

DO

Purchase a Hawaiian Shirt



DO NOT

Use obsoleted APIs

DO NOT

Use DataPages or Themes NuGets

DO

Use .NET Standard Libraries

Evergreen .NET MAUI

What’s my new experience in .NET MAUI?

Building UI in .NET 6

All the XAML and MVVM you know and love

```xaml
<StackLayout>
    <Label Text="Welcome to Maui!" />

    <Button Text="{Binding Text}"
            Clicked="{Binding ClickCommand}" />
</StackLayout>
```

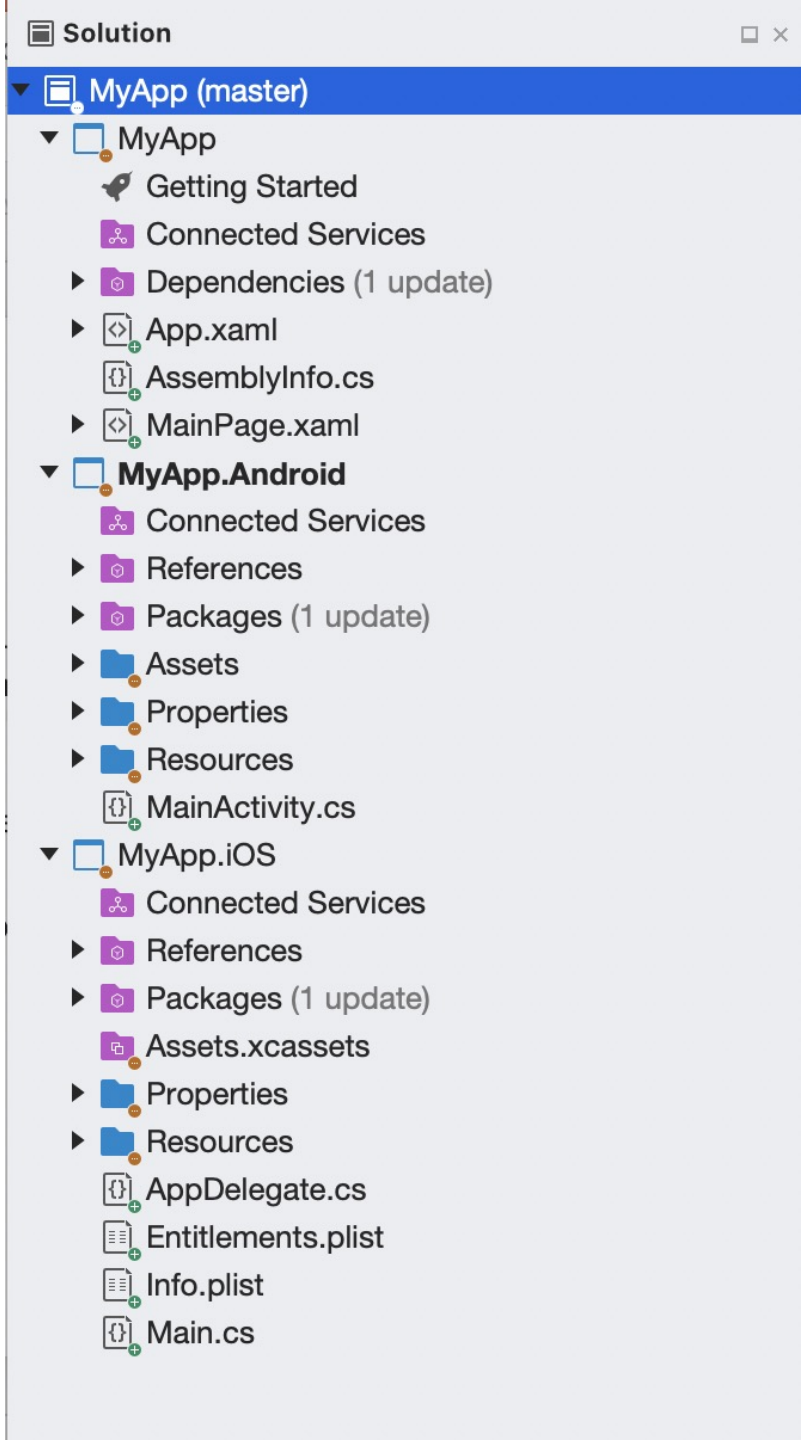
```
0 references
public ICommand ClickCommand { get; }
1 reference
public string Text { get; set; } = "Click me";

int count = 0;
0 references
void ExecuteClickCommand()
{
    count++;
    Text = $"You clicked {count} times.";
}
```

Single Project

Single project alleviates several points of friction when developing apps that target multiple platforms.

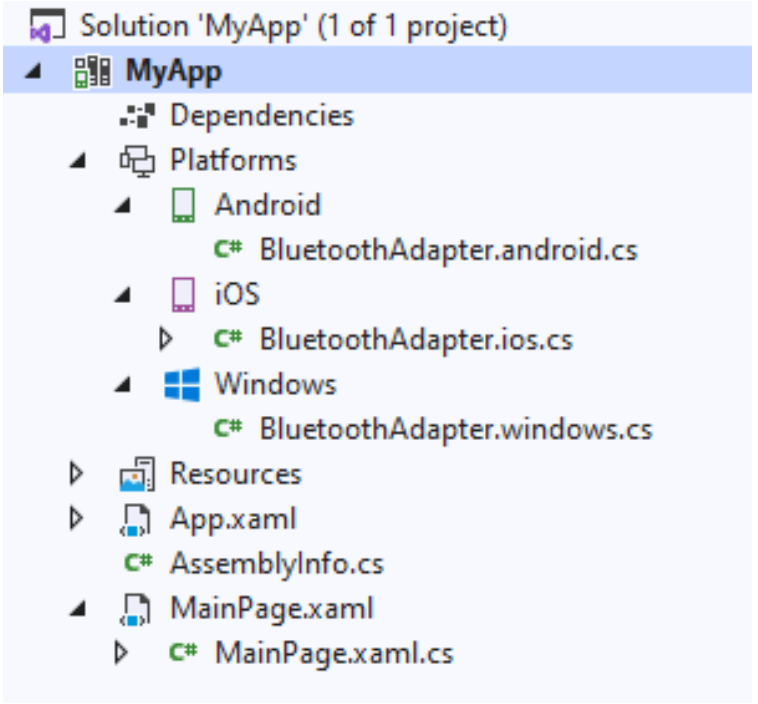
- Images
- Fonts
- Platform code



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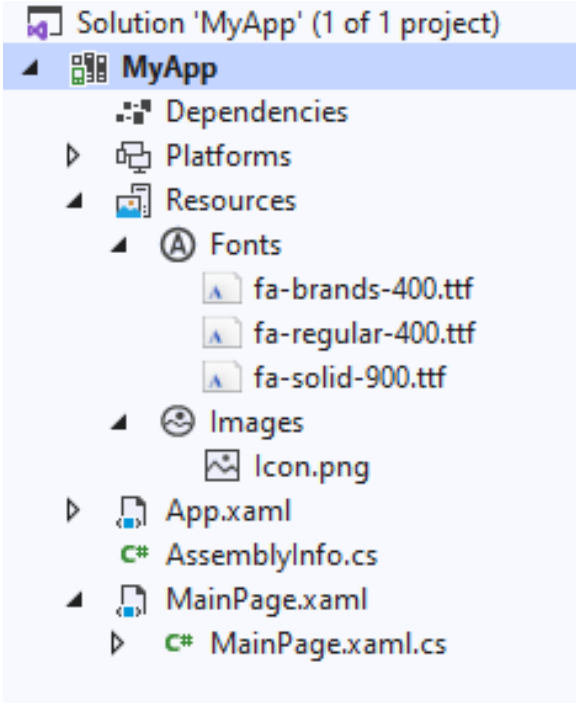
- Multi-targeting
- Images
- Fonts
- Platform code



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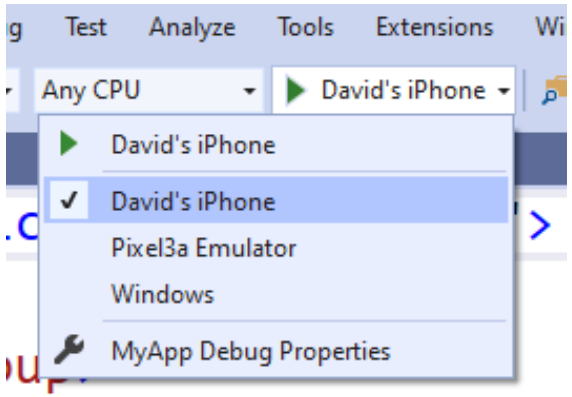
- Images
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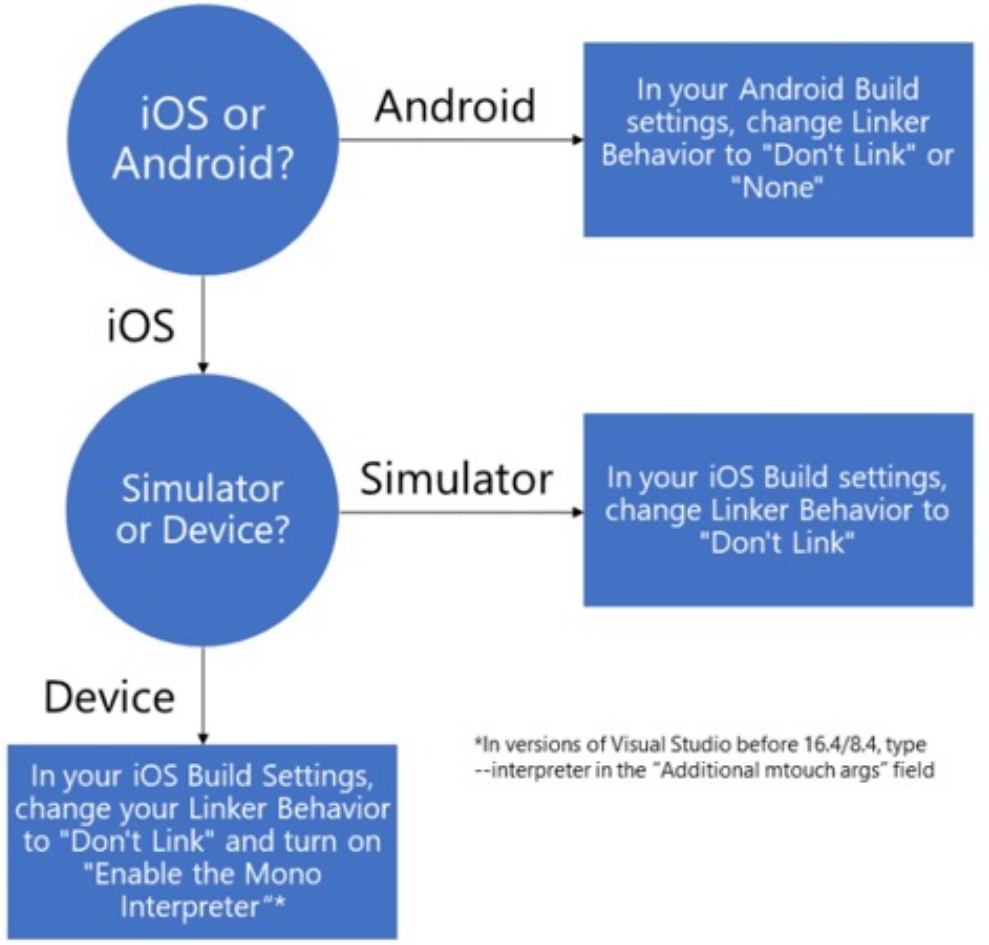
- Images
- Fonts
- Platform code



Hot Reload

Iterate on your application without time consuming rebuilds.

- Android, iOS, macOS, and Windows
- XAML and C#
- State retention
- Multi-devices
- No debugger required



Hot Restart

Extends Hot Reload experience to resources and references.

- iOS from Windows
- Android from macOS or Windows

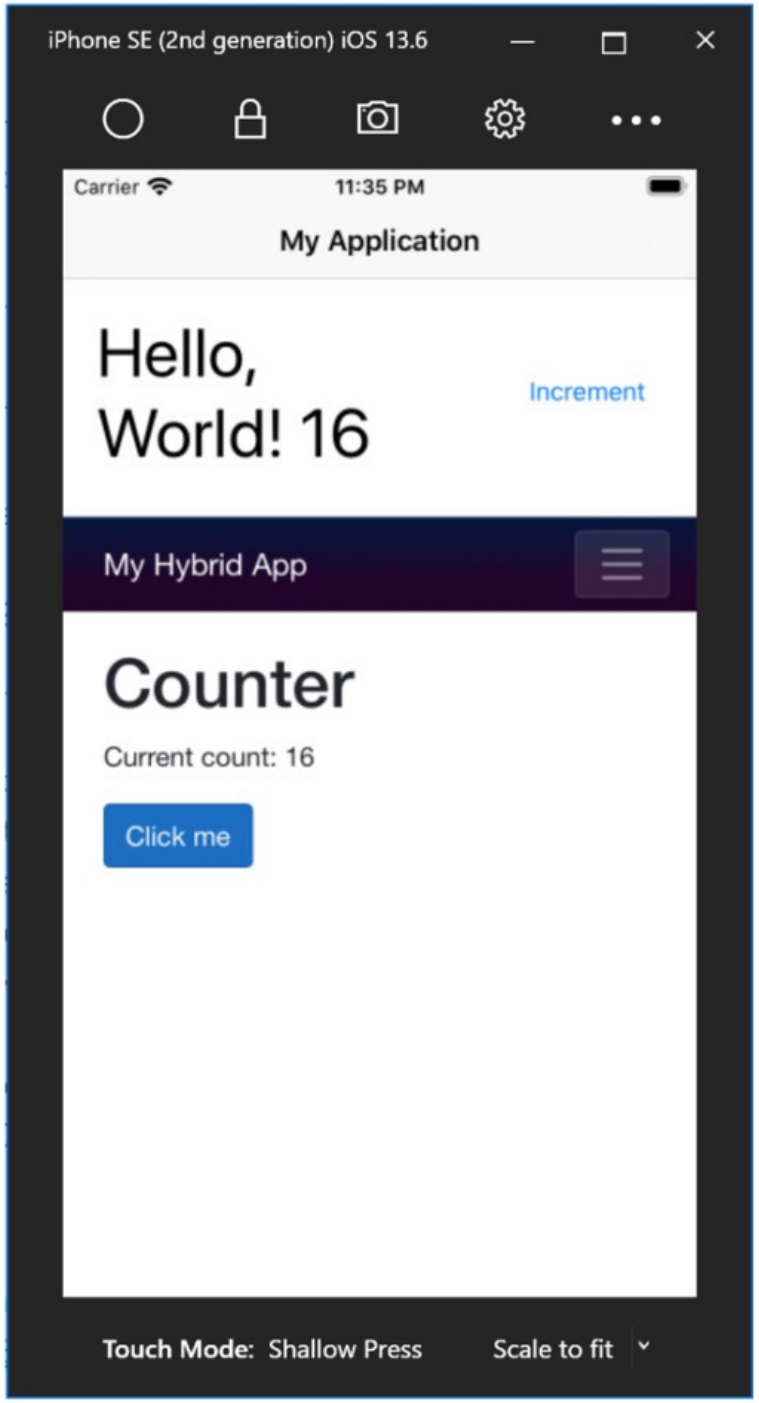


Experiments

What experiences can we build on top of these performant, native APIs?

Blazor

- Razor + .NET MAUI controls
- Hybrid mode
- Visual Studio Code
- Debugging

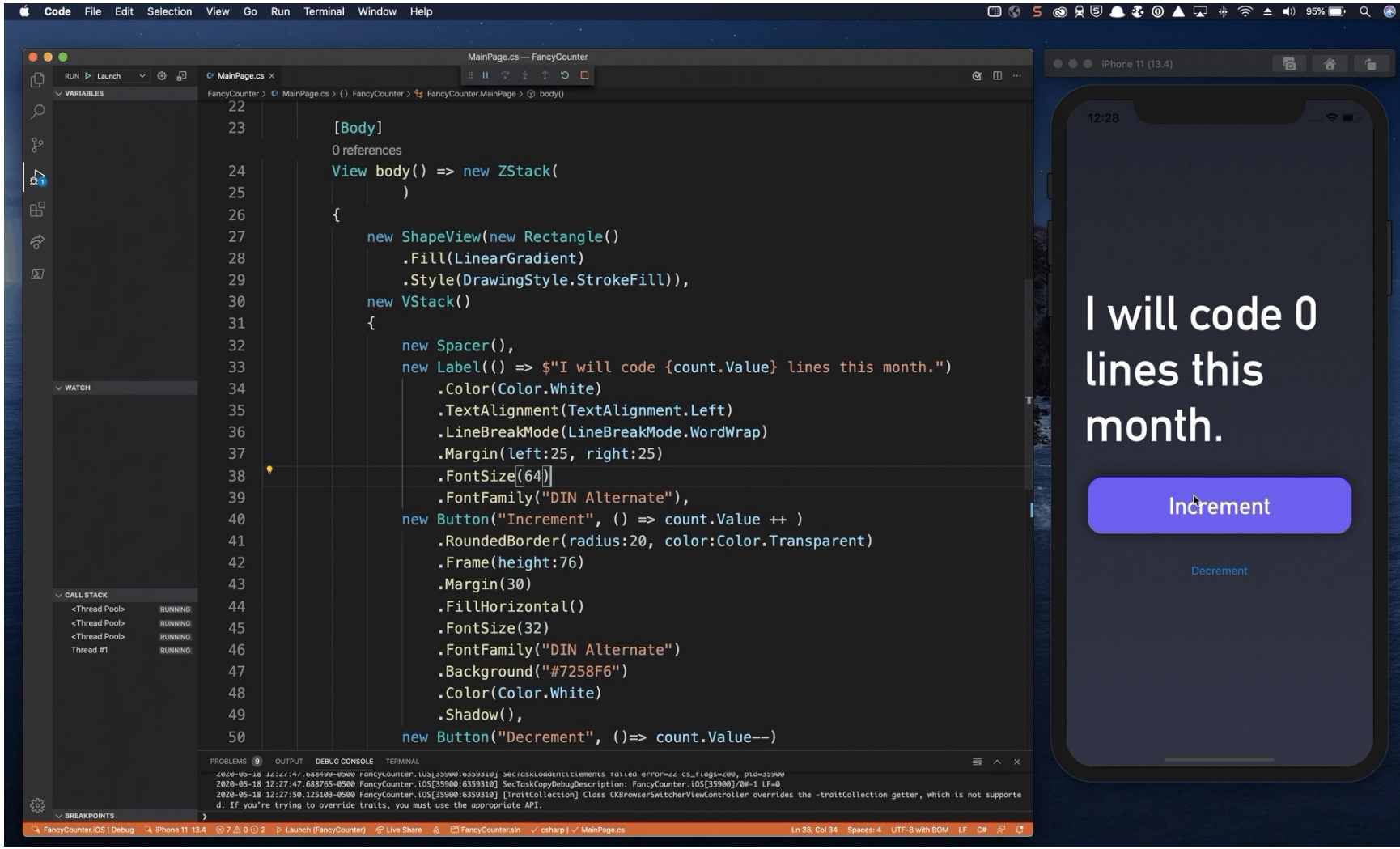


Native UI

Web UI

Comet

- Unidirectional flow of state
- C# fluent syntax
- Focused hot reload
- Visual Studio Code
- Debugging



FAQs

Questions, worries, and the hunt for the cheese.

Should I use Xamarin.Forms now, or wait for .NET MAUI?

Xamarin.Forms. Your project will upgrade to .NET MAUI.

Will you migrate my solution to “Single Project”?

No, your existing projects will continue to work without Single Project. We will migrate projects in place to the .NET 6 common project system.

Documentation will be provided if you wish to manually adopt Single Project.

Do I need to rewrite custom renderers?

No. We are making sure renderers still work.

You will want to migrate them at your convenience.

Will 3rd party UI libraries continue to work?

Yes! At a minimum they will need to recompile adding a target for .NET 6 and switch namespaces.

Will .NET MAUI run on Linux?

Linux is not a supported platform in this version of .NET MAUI.
There is an unofficial backend implemented with GTK.

What “flavor” of XAML will .NET MAUI use?

The same XAML you use in Xamarin.Forms today.

We will revisit some naming changes later in the preview.

Where can I follow progress?

github.com/dotnet/maui

Refreshed regularly from xamarin/xamarin.forms

Thanks for joining!

